

JADE OS · Sales Engineering Execution Plan

8 Phases · 43 substeps · ~116h budget (~14.5 working days) · progress 0.0% (0/43) · generated June 07, 2026 00:00 UTC

#	Phase	Duration	Owner	Steps	Hours	Linked Lab
1	Market Research & Company Identification	3–4 days	Analyst + Founder	0/7	30h	OP-01
2	AI Agent Capability Mapping & Feature Definition	3 days	Engineer + Founder	0/6	20h	OP-03
3	Financial Modeling & ROI Quantification	4 days	Founder + Engineer	0/6	11h	OP-02
4	Detailed Target Company List & Segmentation	2 days	Analyst + Founder	0/5	11h	OP-06
5	Benefits & Features Deep-Dive Document	2–3 days	Founder + Engineer	0/6	17h	OP-05
6	Sales Collateral & Pitch Deck Development	2 days	Founder	0/5	11h	OP-04
7	Go-to-Market Strategy & Outreach Plan	2 days	Founder	0/4	10h	—
8	Reporting, Documentation & Continuous Refinement	~2 days	Founder + Engineer	0/4	6h	—

Phase 1 · Market Research & Company Identification

Duration 3–4 days · Owner Analyst + Founder · 0/7 done · 30h budget · Linked Lab OP-01

Outcome: A clean, source-traceable target universe of 50-150 MN freight companies + 3-4 buyer-persona definitions with documented workflow gaps.

■ 1.1 · Query MN SOS SOFI for freight/trucking/logistics LLCs + Corps with ≥5 employees on SIC 4213/4214/4215/4221/7359

DELIVERABLE · CSV export · ≥500 raw rows

EXIT · rows have legal_name + entity_type + status=active + filing_date

4h · Analyst · → OP-06 · tools: MN SOS SOFI (sos.state.mn.us) · Python csv module

■ 1.2 · Cross-reference exported list with Trucking Association of MN + MN Motor Carriers Association rosters

DELIVERABLE · Joined CSV · de-duped on legal_name + DOT#

EXIT · ≥250 carriers survive de-dup, each row carries TAM_member=true/false

4h · Analyst · → OP-06 · tools: mntruck.org · FMCSA SAFER · openrefine · depends: 1.1

■ 1.3 · Enrich with fleet size, service type, annual revenue from Hoover's + state filings + carrier websites

DELIVERABLE · Enriched CSV with fleet_size, service_type, revenue_range

EXIT · ≥80% of rows have fleet_size populated; revenue is a range, never a fabricated point estimate

8h · Analyst · → OP-06 · tools: D&B; Hoover's · FMCSA QCMobile · company websites · depends: 1.2

■ 1.4 · Score every row for mid-market fit (50–500 vehicles + Minnesota domiciled) and tag mid-market=true

DELIVERABLE · Filtered CSV · mid-market candidates only · 100–250 rows expected

EXIT · every retained row has fleet_size IN [50, 500] AND state==MN

2h · Analyst · → OP-06 · tools: Excel/Python pandas · depends: 1.3

■ 1.5 · Pull 3–5 industry benchmark reports (ATA Trucking Profile, ATRI Operational Costs, BLS OEWS 53-3032, EIA diesel, FMCSA enforcement)

DELIVERABLE · /research/benchmarks/ folder · 5 PDFs + 1 index.md summarizing key numbers

EXIT · every benchmark has: source URL, publication date, the 3 numbers we will cite, the cite-ready quote

6h · Analyst · → OP-01 · tools: ATA · ATRI · BLS OEWS · EIA STEO · FMCSA

■ 1.6 · Lock baseline costs (driver \$/hr, fuel \$/gal, compliance violation cost, dispatcher hours/load) into ops_workbench bench dict

DELIVERABLE · Code · industry_benchmarks dict in build_roi_model already wired

EXIT · Every constant has a comment-citation pointing at the source document in /research/benchmarks/

2h · Engineer · → OP-02 · tools: ops_workbench.py · depends: 1.5

■ 1.7 · Define 3–4 buyer personas (Small Regional · Mid-Market · Intermodal · Specialized/Hazmat) with documented workflow gaps

DELIVERABLE · PITCH_DECK.fact_sheets + collateral persona doc

EXIT · Each persona has: headline · 3 pains · lead_agent · expected ROI range · contact channel preference

4h · Founder · → OP-04 · tools: ops_workbench.PITCH_DECK.fact_sheets · depends: 1.4 · 1.5

Phase 2 · AI Agent Capability Mapping & Feature Definition

Duration 3 days · Owner Engineer + Founder · 0/6 done · 20h budget · Linked Lab OP-03

Outcome: A 6-module agent stack, each with autonomy level, quantified KPI band, decision logic, and the exact input/output contract that the rest of the org can build to.

■ **2.1** · Architect the 6 agent modules · Dispatch · Route+Fuel · Compliance · Pricing · Retention · Maintenance

DELIVERABLE · `AI_ARCHITECTURE.modules` in `ops_workbench.py`

EXIT · Each module documents: id · name · inputs · outputs · decision_logic · KPI band · autonomy

4h · Engineer · → OP-03 · tools: `ops_workbench.py`

■ **2.2** · Assign autonomy levels L1 (suggest) / L2 (act with approval) / L3 (act autonomously, audit only)

DELIVERABLE · Per-module autonomy field with rationale in code comment

EXIT · L3 reserved for low-risk reversible actions; L2 default; L1 for pricing-floor and compliance-block scenarios

2h · Founder · → OP-03 · tools: `ops_workbench.AI_ARCHITECTURE` · depends: 2.1

■ **2.3** · Quantify features (empty-mile 8-15% · fuel 10-18% · violation 40-60% · retention +6-12pp · downtime -20-30%)

DELIVERABLE · Per-module KPI band stored on `AI_ARCHITECTURE.modules[*].kpi`

EXIT · Every KPI carries a benchmark source ID from Phase 1 step 1.5

3h · Engineer · → OP-03 · tools: ATA · ATRI · FMCSA · depends: 1.5 · 2.1

■ **2.4** · Design data pipeline · ELD + TMS + fuel card + insurance + telematics + DOT 511

DELIVERABLE · `AI_ARCHITECTURE.data_pipeline` · 6 stages (Sources · Ingest · Store · Decide · Act · Observe)

EXIT · Each stage names concrete vendors/APIs; no abstract placeholders

4h · Engineer · → OP-03 · tools: `AI_ARCHITECTURE.data_pipeline` · depends: 2.1

■ **2.5** · Pre-map APIs for top 3 MN TMS · Descartes · McLeod · TMW (or Selerant for hazmat heavy)

DELIVERABLE · `/research/tms_apis/` · 3 markdown docs · auth model + endpoints + rate limits + sandbox availability

EXIT · Each TMS has a concrete OAuth or API-key recipe + a tested ping example

4h · Engineer · → OP-03 · tools: TMS vendor docs · Postman · depends: 2.4

■ **2.6** · Build feature-to-pain-point 2×2 matrix · quantified per persona × per agent

DELIVERABLE · `PITCH_DECK.fact_sheets` + a `/research/feature_matrix.csv`

EXIT · Every cell carries a numeric impact range traceable to a benchmark

3h · Founder · → OP-04 · tools: Excel · `PITCH_DECK` · depends: 1.7 · 2.3

Phase 3 · Financial Modeling & ROI Quantification

Duration 4 days · Owner Founder + Engineer · 0/6 done · 11h budget · Linked Lab OP-02

Outcome: A live, recompute-on-demand ROI model with 3 archetypes, 6 savings categories per archetype, 3-yr NPV, payback months, and $\pm 10\%$ sensitivity — every number is benchmark-cited.

■ 3.1 · Define 3 archetypes · Small Regional 25-75 · Mid-Market 100-250 · Specialized/Hazmat 50-150

DELIVERABLE · `arch_assumptions` dict in `build_roi_model` with fleet defaults + `dispatch/fuel/compliance/retention/downtime/claims` pct uplifts

EXIT · Each archetype label + fleet default + 6 percentage assumptions visible in code

2h · Engineer · → OP-02 · tools: `ops_workbench.build_roi_model`

■ 3.2 · Codify 6 cost-savings categories per archetype · labor · fuel · compliance · downtime · retention · claims

DELIVERABLE · `by_category_usd` output keys returned by `/api/workbench/labs/op-02/run`

EXIT · Every category has a formula visible in `build_roi_model`; no fudge factors

2h · Engineer · → OP-02 · tools: `ops_workbench.build_roi_model` · depends: 3.1

■ 3.3 · Compute 3-year NPV @ 10% discount + payback in months

DELIVERABLE · `three_year` output: `net_y1/y2/y3` + `npv_at_10pct` + `payback_months`

EXIT · Math reproducible by hand from inputs; matches the recompute UI exactly

2h · Engineer · → OP-02 · tools: `ops_workbench.build_roi_model` · depends: 3.2

■ 3.4 · Sensitivity analysis · $\pm 10\%$ on annual savings (proxy for fuel/wages/hours)

DELIVERABLE · `sensitivity` dict · `conservative` / `base` / `optimistic`

EXIT · Bands shown alongside base every recompute; not hidden in code

1h · Engineer · → OP-02 · tools: `ops_workbench.build_roi_model` · depends: 3.3

■ 3.5 · Scenario switcher · conservative/base/optimistic + archetype + fleet override exposed in UI

DELIVERABLE · `WorkbenchPanel` `LabOP02` + `ConsoleWorkbenchPanel` `RoiPanel` selectors

EXIT · Switch updates 6-category breakdown + NPV + payback live, <300ms

3h · Engineer · → OP-02 · tools: `WorkbenchPanel.jsx` · `ConsoleWorkbenchPanel.jsx` · depends: 3.4

■ 3.6 · Source attribution · model returns `sources[]` every run (ATA · ATRI · BLS · EIA · FMCSA)

DELIVERABLE · `model.sources` array

EXIT · ≥ 5 distinct sources cited; no source listed without an in-formula citation

1h · Founder · → OP-02 · tools: `ops_workbench.build_roi_model` · depends: 3.2

Phase 4 · Detailed Target Company List & Segmentation

Duration 2 days · Owner Analyst + Founder · 0/5 done · 11h budget · Linked Lab OP-06

Outcome: A 15-25 row Minnesota target list with fleet, service, revenue range, TMS in use, AI-readiness score, persona, sales stage, and recommended contact method.

■ 4.1 · Pull the curated MN_FREIGHT_SEED list from fmcsa_lookup.py into db.prospects (idempotent)

DELIVERABLE · ≥ 14 real companies upserted · all is_synthetic=false · DOT/MC populated where known

EXIT · Each prospect has verification_source set

1h · Engineer · → OP-06 · tools: fmcsa_lookup.MN_FREIGHT_SEED

■ 4.2 · Enrich with TMS in use (best-effort from public sources)

DELIVERABLE · tms_vendor column populated where discoverable; left null where not — never guessed

EXIT · Null cells outnumber confidently-set cells until verified

4h · Analyst · → OP-06 · tools: LinkedIn · carrier websites · industry pubs · depends: 4.1

■ 4.3 · Score AI readiness 0-5 per prospect (TMS modernity · ELD vendor · IT bench · exec sponsor signal)

DELIVERABLE · ai_readiness column on db.prospects

EXIT · Every score has a 1-line rationale in notes

3h · Founder · → OP-06 · tools: Custom rubric in /research/ai_readiness.md · depends: 4.2

■ 4.4 · Segment by persona × sales stage (cold / warm / hot / pilot-ready)

DELIVERABLE · persona + sales_stage columns on db.prospects

EXIT · Top-of-funnel $\leq 30\%$ in hot/pilot-ready; cohort split visible in dashboard

2h · Founder · → OP-06 · tools: DesignPartnersPanel · depends: 4.3

■ 4.5 · Recommend contact method per row (cold call · LinkedIn warm intro · referral · event)

DELIVERABLE · contact_method column

EXIT · Method matches sales_stage rule book (cold → cold call/LinkedIn; warm → referral)

1h · Founder · → OP-06 · tools: Operator rule book · depends: 4.4

Phase 5 · Benefits & Features Deep-Dive Document

Duration 2–3 days · Owner Founder + Engineer · 0/6 done · 17h budget · Linked Lab OP-05

Outcome: A polished 20-30 page operator-grade technical brief PDF that a CFO can read end-to-end with every claim benchmark-traceable.

■ 5.1 · Author the executive overview (1 page · 5 headline bullets)

DELIVERABLE · TECHNICAL_DOC_SECTIONS[0] in ops_workbench.py

EXIT · Reads to a non-technical exec in <2 minutes

2h · Founder · → OP-05 · tools: ops_workbench.TECHNICAL_DOC_SECTIONS

■ 5.2 · Six module deep-dives · ~3 pages each (M1-M6)

DELIVERABLE · TECHNICAL_DOC_SECTIONS[1..6]

EXIT · Each section covers inputs · decision · action · KPI · failure modes

8h · Engineer · → OP-05 · tools: AI_ARCHITECTURE.modules · depends: 2.1

■ 5.3 · ROI analysis · 2 illustrative case studies (Small Regional 50t · Mid-Market 175t)

DELIVERABLE · TECHNICAL_DOC_SECTIONS[7]

EXIT · Numbers match build_roi_model output to the dollar

2h · Engineer · → OP-05 · tools: build_roi_model · depends: 3.3 · 5.2

■ 5.4 · Implementation roadmap · 4 phases (Readiness · Baseline · Pilot · Scale)

DELIVERABLE · TECHNICAL_DOC_SECTIONS[8]

EXIT · Every phase has a duration, exit criteria, and operator action

2h · Founder · → OP-05 · tools: roadmap_template

■ 5.5 · Minnesota opportunity snapshot + technical appendix (APIs · auth · audit · LLM substrate)

DELIVERABLE · TECHNICAL_DOC_SECTIONS[9]

EXIT · API table mirrors AI_ARCHITECTURE.api_surface exactly

2h · Engineer · → OP-05 · tools: AI_ARCHITECTURE.api_surface · depends: 2.4

■ 5.6 · Render to PDF via reportlab (`generate\_technical\_doc\_pdf`)

DELIVERABLE · PDF file at /app/backend/static_workbench/op05_*.pdf

EXIT · First 4 bytes = %PDF, size >5KB, opens in any PDF reader

1h · Engineer · → OP-05 · tools: ops_workbench.generate_technical_doc_pdf · depends: 5.1 · 5.2 · 5.3 · 5.4 · 5.5

Phase 6 · Sales Collateral & Pitch Deck Development

Duration 2 days · Owner Founder · 0/5 done · 11h budget · Linked Lab OP-04

Outcome: A 12-slide pitch deck PDF + 4 persona fact sheets + a Readiness Assessment + a competitive brief — ready for live discovery calls.

■ **6.1** · Compose the 12-slide pitch (problem · solution · 6 modules · ROI · 2 case profiles · competitive · pricing · ask)

DELIVERABLE · `PITCH_DECK.slides` in `ops_workbench.py`

EXIT · Slide count == 12, every slide has a title + (subtitle | bullets)

4h · Founder · → OP-04 · tools: `ops_workbench.PITCH_DECK` · depends: 1.7 · 5.2

■ **6.2** · 4 persona fact sheets · one per buyer persona · headline + 3 pains + lead agent + ROI

DELIVERABLE · `PITCH_DECK.fact_sheets[0..3]`

EXIT · Every fact sheet has a quantified ROI range from `build_roi_model`

2h · Founder · → OP-04 · tools: `PITCH_DECK.fact_sheets` · depends: 1.7 · 3.5

■ **6.3** · Readiness Assessment workbook · 3 areas (Data · Process · Tech & Org) · 4-5 questions each

DELIVERABLE · `PITCH_DECK.readiness_assessment`

EXIT · Yes/no answer model; takes a prospect <15 min to complete

2h · Founder · → OP-04 · tools: `PITCH_DECK.readiness_assessment`

■ **6.4** · Competitive brief vs Descartes / Omnitrac / Geotab / internal-build

DELIVERABLE · `PITCH_DECK.competitive_brief`

EXIT · Each comp has a 2-sentence positioning statement and an explicit ABOVE-the-stack framing

2h · Founder · → OP-04 · tools: `PITCH_DECK.competitive_brief`

■ **6.5** · Render deck to PDF (`generate_pitch_deck_pdf`) · landscape · brand fonts

DELIVERABLE · `/app/backend/static_workbench/op04_*.pdf`

EXIT · Deck PDF >8KB, 12+ pages, opens cleanly

1h · Engineer · → OP-04 · tools: `ops_workbench.generate_pitch_deck_pdf` · depends: 6.1

Phase 7 · Go-to-Market Strategy & Outreach Plan

Duration 2 days · Owner Founder · 0/4 done · 10h budget

Outcome: Channel mix per persona, a 30/60/90 outreach sequence per persona, and a signed pilot agreement template with success metrics declared up-front.

■ **7.1** · Channel mix · direct vs partner per persona (Small Regional → direct; Specialized → direct + insurance-broker partner; Mid-Market → hybrid)

DELIVERABLE · `/research/gtm_channel_mix.md`

EXIT · Each persona has a primary and a fallback channel with reasoning

2h · Founder · tools: Operator playbook · depends: 1.7

■ **7.2** · 30/60/90-day outreach sequence per persona (LinkedIn nudges · executive briefing · ROI workbook share · pilot agreement)

DELIVERABLE · `/research/outreach_sequences.md` · 4 persona × 3 cadence rows

EXIT · Each touch carries a CTA and a success indicator

4h · Founder · tools: LinkedIn Sales Nav · Resend · depends: 7.1

■ **7.3** · Pilot agreement template · 6-month term · \$25-40k all-in · success-metric definition

DELIVERABLE · `/research/pilot_agreement_template.md`

EXIT · Template names ≥3 success metrics tied to KPIs from Phase 2 step 2.3

3h · Founder · tools: legal_review · PITCH_DECK · depends: 2.3 · 3.5

■ **7.4** · Decision checkpoints @ month 4 of pilot · agreed in writing pre-pilot

DELIVERABLE · Pilot template appendix · checkpoint clause

EXIT · Either party can pause/terminate at month 4 if metrics aren't on the trajectory line

1h · Founder · tools: pilot_agreement · depends: 7.3

Phase 8 · Reporting, Documentation & Continuous Refinement

Duration 1–2 days · Owner Founder + Engineer · 0/4 done · 6h budget

Outcome: Weekly lessons-learned cadence + a pipeline + decision-checkpoint dashboard so every prospect interaction informs the next.

■ **8.1** · Weekly lessons-learned log · cadence stand-up · who learned what · what changes for next cycle

DELIVERABLE · `/research/lessons_log.md` · appended every Friday

EXIT · ≥ 3 lessons per week or note 'nothing new' explicitly

1h · Founder · tools: calendar reminder

■ **8.2** · Pipeline reporting dashboard · prospect count by stage $\times$ persona $\times$ week

DELIVERABLE · Existing DesignPartnersPanel + new charts as needed

EXIT · Visible to founder + analyst on a single screen

3h · Engineer · tools: DesignPartnersPanel

■ **8.3** · Decision checkpoint cadence · update `workbench_decisions` monthly · re-flip when evidence changes

DELIVERABLE · PATCH `/api/workbench/decisions/{id}` every month or when a pilot insight contradicts the current choice

EXIT · `decisions.updated_at` within last 30 days for active decisions

1h · Founder · tools: WorkbenchPanel DecisionsTracker

■ **8.4** · Audit trail review · weekly verify the immutable chain via `/api/audit/verify`

DELIVERABLE · Chain `ok=true` every Friday; any failure escalated immediately

EXIT · Failure triggers incident comm + freeze on rate-floor overrides until resolved

1h · Engineer · tools: `audit_trail.verify` · RiskGuardPanel

— end of plan —